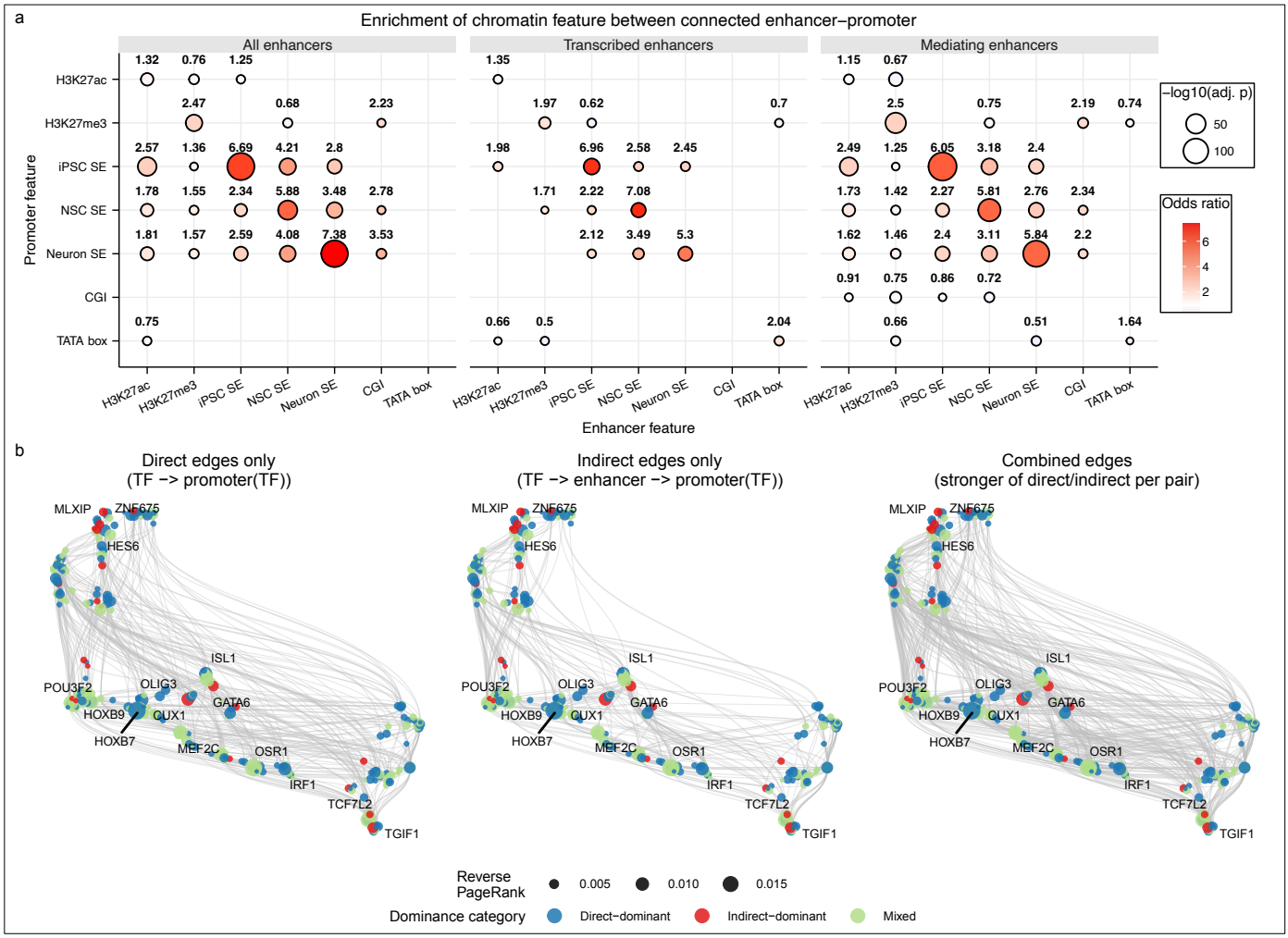


Extended Figure 7 | Epigenomic concordance of enhancer–promoter interactions and TF–TF regulation.

a, Co-enrichment of epigenomic features between enhancers and promoters connected by the final high-confidence GRN edges. Columns and rows represent enhancer- and promoter-associated features, respectively, including H3K27ac, H3K27me3, cell-type-specific super-enhancers (SEs), CpG islands (CGIs) and TATA boxes. Color indicates the odds ratio from a chi-square test of association (red, enrichment; blue, depletion. Panels show interactions involving all enhancers (left), transcribed enhancers (center) and mediating enhancers (right). **b**, UMAP representation of direct and enhancer-mediated indirect TF–TF regulatory relationships in the final high-confidence GRN. Direct relationships were defined by TF–promoter edges targeting another TF gene (left), whereas indirect relationships were defined by TF–enhancer–promoter paths targeting a TF gene (center); the combined network is shown on the right. Node positions are fixed to the UMAP layout of the combined network, and node size represents reverse PageRank. TFs are classified according to the relative contribution of direct and indirect outgoing relationships as direct-dominant (>2/3 direct), indirect-dominant (<1/3 direct), mixed or without outgoing edges.